MILITARY SPECIFICATION

REGULATOR, COMPRESSED AIR, HIGH PRESSURE (3500 PSI)

This specification has been approved
by the Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 This specification covers one type of high pressure, compressed air regulator.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings and publications, of the issue in effect on dates of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)
UU-P-171	paper, Wrapping, Waterproofed Kraft

Military

MIL-A-140	Adhesive, Water-Resistant, Waterproof, Barrier Material
MIL-E-5272(ASG)	Environmental Testing; Aeronautical and Associated Equipment, General Specifications for
MIL-P-7936	Parts and Equipment, Preparation for Delivery
MIL-P-7105	Pipe Threads, Taper, Aeronautical, National Form, symbol ANPT
MIL-S-7742	Screw Threads, Standard Optimum Selected Series; General Specification for
MIL-A-8625(ASG)	Anodic-Coatings, for Aluminum and Aluminum Alloys
MIL-T-9107	Test Reports; Preparation of
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking of Shipments
MIL-STD-130	Identification Markings of U.S. Military Property

FSC-4310

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:23Z
Check the source to verify that this is the current version before use.

DRAWINGS

Air Force - Aeronautical Standard Drawing

AND103988	Metals - Definition of Dissimilar

PUBLICATIONS

Air Force - Navy Aeronautical Bulletins

143	Specifications and Standards; Use of
405	Storage Life - Aeronautical Articles

(When requesting specifications, standards, and other publications refer to both title and number. Copies of this specification and applicable specifications, standards, and other publications may be obtained upon application to the Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania, Attention! Code CDS.)

3. REQUIREMENTS

3.1 Preproduction.- This specification makes provisions for preproduction testing.

3.2 Materials.- Material shall be as specified herein. Material not definitely-specified shall be of a good commercial quality entirely suitable for the purpose. Material shall be-free from all defects-and imperfections that might affect the serviceability of the finished product. Dissimilar metals as defined by Drawing AND10398 shall not be placed in contact with each other unless the adjoining surfaces are suitably coated to preclude any corrosive action.

3.2.2 Nonmetallic Materials.- Any nonmetallic material that deteriorates or is otherwise affected adversely by continued use with high pressure air for the normal service period of one (1) year shall not be used.

3.2.3 Nonferrous Materials.- Nonferrous materials shall be used in the construction of the regulator, except where ferrous materials are essential.

3.2.4 Protective Treatment.- When materials are used in the construction of the regulator that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:23Z
Check the source to verify that this is the current version before use.

3.2.5 Selection of Materials. - Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with ANA Bulletin No. 143, except as provided in the following paragraph.

3.2.5.1 Standard Parts. - Standard parts (MS, AN, or JAN) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc., may be used, provided they possess suitable Properties and are replaceable by the standard parts (MS, AN, or JAN) without alteration, and provided the corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is not suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.3 Design and Construction.- The regulator shall be so designed and constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, installation, and service.

3.3.1 The regulator shall be so constructed that adjustments and repairs can easily be made by the personnel of operating units and overhaul bases without the use of special tools.

3.3.2 The regulator shall be designed and constructed to reduce pressure from an initial source at pressures up to 3750 psi to any delivery pressure between 0 and 3500 psi, with a delivery rate of not less then 25 CFM of air at any combination of inlet and outlet pressures within the design limits. The pressure-adjustment control shall consist of a thread screw provided with a suitable knob type handle which will require a maximum of 30 inch pounds of torque to turn. The final delivery pressure shall not vary more than three percent from the preset value.

3.3.3 Inlet Screen. - A suitable filter, equivalent to a 200 -mesh screen, shall be provided in the inlet connection to prevent foreign particles from entering the regulator, and shall be removable for inspection and cleaning purposes.

3.3.4 Pressure Gages.- The regulator shall be provided with two commercial-type 2 1/2 inch pressure gages. One gage shall indicate inlet pressures from 0 to 4000 psi, and the other shall indicate delivery pressures from 0 to 4000 psi. The accuracy for each gage shall be within the commercial tolerances established for this type of equipment, however, the tolerance shall not exceed 2 percent of the delivery pressure indicated.

3.3.5 Safety Valve. - A resealing-type safety valve shall be provided in the delivery side of the regulator to prevent excessive build-up of pressure in this chamber. The safety valve shall be preset and sealed to prevent alteration of venting pressure. Venting shall begin at a delivery pressure in excess of 3600 psi, and the safety valve shall reseal at a pressure of not less than 3400 psi.

Source: https://assist.dla.mil -- Downloaded: 2026l0-19T15:23Z
Check the source to verify that this is the currentversion before use.

3.3.6 The regulator specified herein shall operate satisfactorily in any position in which it is mounted.

3.3.7 The inlet and outlet of the regulator shall be clearly marked.

3.4 Interchangeability. - All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturers part numbers shall be governed by the drawing number requirements of Specification MIL-D-70327.

3.5 Threads. -

3.5.1 Screw. - Unless otherwise specified, all machine screw threads shall conform to Specification MIL-S-7742, Class 2 fit.

3.5.2 Pipe. - Pipe threads shall conform to Specification MIL-P-7105.

3.6 Dimensions. - The maximum over-all dimensions of the regulator shall be 4" diameter or 4" x 4", and 12" long less gages.

3.7 Mounting Bolt Holes. - Four mounting bolts or tapped holes shall be furnished. The bolt holes or tapped-hole depth shall be such that the regulator may be mounted on panels from 1/8 to 1/2 inch in thickness.

3.8 Weight. - The weight of the completed regulator with gages shall not exceed 25 pounds.

3.9 Finishes and Protective Coatings. - All protected metal parts and surfaces that are constructed from corrosion resistant material and on which a protective coating would be detrimental to proper functioning, need not be given a protective coating. All exposed metal parts and surfaces, and all protected metal parts on which a protective coating will not be detrimental to proper functioning shall be treated to satisfy the requirements of this specification.

3.9.1 Magnesium Alloy Parts. - Magnesium base alloy parts shall not be used.

3.9.2 Aluminum Alloy Parts. - Aluminum alloy surfaces shall be anodized in accordance with Specification MIL-A-8625(ASG).

3.9.3 Ferrous Parts. - All ferrous parts shall be cadmium plated in accordance with Specification QQ-P-416, Type II or III, as applicable, and of a class that is adequate to achieve the degree of protection required.

3.10 Performance. - The regulator shall be capable of operating under the following conditions:

a. The regulator shall not sustain hum or vibration within its operating range of pressure and flow.

b. Temperature: The regulator shall be capable of operating herein within the temperature of -54° to ± 71° C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:23Z
Check the source to verify that this is the current version before use.

c. Sand and dust: Exposure to airborne sand and dust particles encountered in normal and desert operations.

d. Freezing rain: Exposure to rain at freezing or below freezing temperatures.

e. Vibration: The regulator shall be capable of withstanding vibration having a double amplitude from 0.00 to 0.125 inch at a continuously varying frequency from 600 to 1000 cfm.

f. Seasoning: The regulator shall be capable of being cycled 500 times between 500 and 2800 psi gage on the outlet side when the inlet side of the regulator is connected to a pressure of 3500 psi gage.

3.11 Identification of Product. - Equipment, assemblies and parts shall be marked for identification in accordance with Standard MIL-STD-130.

3.12 Workmanship. -

3.12.1 General. - The regulator, including all parts and accessories, shall be fabricated and and finished in a thoroughly workmanship manner. particular attention shall be given to freedom from blemishes, defects, burrs, sharp edges, and accuracy of dimensions. Workmanship shall be in accordance with high-grade commercial practice covering this type of equipment.

3.12.2 Cleaning.- All parts of the regulator shall be thoroughly cleaned and kept free of oil or other inflammable compounds.

4. QUALITY ASSURANCE PROVISIONS

The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure suppliers and services conform to prescribed requirements.

4.1 Classification of Tests. - The inspection and testing of the regulators shall be classified as follows:

4.1.1 Preproduction Tests. - Preproduction tests are those tests accomplished on a sample representative of the production regulators, to determine that the production regulators meet the requirements of this specification. All tests shall be conducted on the same sample.

4.1.2 Inspection Tests. - Inspection tests are those tests performed on individual lots which have been submitted for acceptance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:23Z
Check the source to verify that this is the current version before use.

4.1.3 Preproduction Test Report. - After the contractor completes the preproduction tests, he shall prepare a preproduction test report in accordance with Specification MIL-T-9107 and furnish three (3) copies of the report to the procuring activity.

4.2 Test Condition. - Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches HG) and at room temperature (approximately 25°C.). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition. The gas to be used to conduct the tests shall be compressed air.

4.3 Preproduction Tests. - Preproduction tests shall be conducted on one regulator. These tests shall consist of all the tests specified under Inspection Tests and, in addition, the following tests.

4.3.1 Low Temperature. - The regulator shall be subjected to a temperature of -54°C. for 3 hours. After the 3 hour period and while still at the specified temperature, the regulator shall be subjected to the test specified in paragraph 4.4.1.3.

4.3.2 High Temperature. - The regulator shall be connected to an inlet pressure of at least 3000 psi gage, and shall be subjected to a temperature of ± 71°C. for a period of 7 hours. After the 7 hour period, the regulator shall be brought to room temperature and subjected to the test specified in paragraph 4.4.1.3.

4.3.3 Vibration. - The regulator shall be subjected to vibration for a period of 3 hours as follows:

4.3.3.1 The vibration shall be in a vertical plane with the regulator mounted at an angle of 45 degrees from the horizontal; it shall have a double amplitude of from 0.00 to 0.125 inch, and continuously varied frequency of from 600 to 1000 cpm. After the 3 hour period, vibration shall be stopped and the regulator subjected to the test specified in paragraph 4.4.1.3.

4.3.4 Seasoning. - With the regulator connected to an inlet pressure of 3500 psi gage, the regulator shall be cycled between 500 and 2800 psi gage on the outlet gage for at least 500 cycles. The regulator shall then be subjected to the test specified in paragraph 4.4.1.3.

4.3.5 Sand and Dust. -The regulator, with gages attached and inlet and outlet openings tightly closed, shall be subjected to the Sand and Dust Test in accordance with Procedure I, of Specification MIL-E-5272. Upon completion of the exposure in the dust chamber, the regulator and gages shall be tested for proper operation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:23Z
Check the source to verify that this is the current version before use.

4.3.6 Freezing Rain. - The regulator with gages attached and inlet and outlet openings tightly closed, shall be subjected to a rain test at freezing temperatures (equivalent to an outdoor freezing rain), or a rain test followed by chilling to below freezing before the unit can dry appreciably. While still below freezing temperatures, the regulator shall be subjected to the test specified in paragraph 4.4.1.3.

4.4 Inspection Tests. - Inspection tests shall consist of Individual Tests and Sampling Tests.

4.4.1 Individual Tests. - Each regulator shall be subjected to the following tests.

4.4.1.1 Examination of Product. - Each regulator shall be inspected to determine compliance with the requirements specified herein with respect to material, workmanship, and marking.

4.4.1.2 The regulator shall be connected to an inlet pressure of 3500 psi gage, and with a delivery pressure of 500 psi, the flow rate of the regulator shall be measured. This flow rate shall be at least 25 cfm.

4.4.2 Sampling Tests. - A random sample shall be selected from each lot of regulators in accordance with MIL-STD-105, Inspection Level L-5, with Acceptance Number zero, and be subjected to the following test:

4.4.2.1 Safety Valve. - The safety valve shall be tested to determine that the venting starts at a delivery pressure of not less than 3600 psi, and shall be tested to indicate that it reseals after venting at a pressure of not more than 3400 psi.

4.4.3 Rejection and Retest. - Regulators which have been rejected may be reworked or have parts replaced to correct the defects, and resubmitted for acceptance. However, final acceptance of the items on hand and subsequent production is contingent upon the Inspector's decision regarding the over-all conformance of the product to specification requirements. When corrective action has been accomplished, all necessary tests shall be repeated. If investigation indicates that the defects may exist in items previously accepted, full particulars concerning the defects, including recommendations for caution, will be furnished to the contracting officer.

4.4.3.1 When sampling tests are specified on a number of items that are selected from a production run and one or more of this number fails to meet the specified tests, acceptance of all items still on hand and subsequent production will be withheld until the extent and cause of failure is determined. For operational reasons, individual tests may be continued pending investigation of a sampling test failure.

4.5 All parts, specimens, or assemblies destroyed in making tests required by this specification, to determine compliance with the specification, shall be in addition to the quantity specified in the contract and shall be furnished without increasing the cost of the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:23Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Application. - The requirements of Section 5 apply only to direct purchases by or direct shipments to the Government.

5.2 Preservation. - The use of preservation compounds is not required for this equipment in preparing for shipment and storage.

5.3 Packaging and Packing. - The regulator shall be packaged and packed for shipment in accordance with Specification MIL-P-7936 Level A, B, or C, (see 6.3).

5.4 Marking of Shipments. - Interior packages and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129.

5.4.1 Reinspection Dates. - Unless otherwise specified, reinspection markings shall be in accordance with ANA Bulletin No. 405.

6. NOTES

6.1 Intended Use. - The high pressure air regulator covered by this specification is for use on the high pressure side of portable air compressors to regulate the delivery air of the compressors.

6.2 Ordering Data. - Procurement documents should specify the following:

a. Title, number and date of this specification

b. Selection of applicable levels of preservation, packing and packaging (See Section 5).

c. Where the preproduction test sample should be sent and instructions concerning the submittal of the test report (See 4.1.3).

NOTICE : When government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the government may have formulated, furnished or in any way supplied, the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T15:23Z
Check the source to verify that this is the current version before use.